

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination-2023

Course: B. Tech.

Branch : Electrical & Allied Engineering

Subject Code & Name: BTEEC702 High Voltage Engineering

Semester :VII

Max Marks: 60

Date:04-1-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain properties of solid insulators in detail.	CO4	6
B) Explain theories of breakdown in liquids.	CO2	6
C) Explain breakdown in non-uniform fields and corona discharge.	CO4	6
Q.2 Solve Any Two of the following.		12
A) Explain Townsend's current growth equation.	CO1	6
B) Explain Streamers theory and Paschen's law in detail.	CO5	6
C) Explain breakdown in non-uniform fields and corona discharge, corona under positive & negative polarities in detail.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain theories of breakdown in liquids.	CO6	6
B) Explain System control of over voltage due to switching protection of transmission line against over voltage.	CO2	6
C) Describe propagation of lightning voltage & current waves on transmission lines.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Write a short note on:	CO4	6
1. Insulation co-ordination.		
2. surge diverters		
B) Describe multistage impulse generator and marx circuit.	CO2	6
C) Write a short note on dielectric constant & loss factor.	CO3	6
Q. 5 Solve Any Two of the following.		12
A) Explain non- destructive testing used for high voltage.	CO1	6

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| B) Explain high voltage tests on electrical appliances & power apparatus & electrical motors. | CO2 | 6 |
| C) Explain radio interference measurement. | CO6 | 6 |

***** End *****